

Configuring Cloud Logging & Security Monitoring

Skills Gained

- Cloud Security Monitoring & Visibility
- Centralized Logging using Google Cloud Logging
- Log Analysis with Logs Explorer
- Resource-level log filtering (Compute Engine VMs)
- Severity-based event prioritization
- Security incident investigation using log metadata
- Understanding SOC-style log investigation workflows
- Practical hands-on experience with Google Cloud Platform (GCP)

Project Objective

The objective of this project was to build hands-on experience with **cloud security monitoring and logging**, focusing on how security teams analyze cloud-generated logs to detect, investigate, and prioritize potential incidents.

Specifically, this project aimed to:

- Analyze cloud-generated logs using Google Cloud Logging
- Identify security-relevant events from collected logs
- Filter logs by resource type (VMs) and severity levels
- Inspect detailed log entries to simulate real-world incident investigation

Tools & Technologies Used

- Google Cloud Platform (GCP)
- Google Cloud Logging

- **Logs Explorer**

Step 1: Cloud Logging Environment Access

Google Cloud Logging was accessed through the Google Cloud Console. Logging in GCP is enabled by default, allowing immediate visibility into activity across cloud resources without additional configuration.

This step confirmed that the cloud environment was actively generating and collecting logs from multiple services.

Step 2: Accessing Logs Explorer

Logs Explorer was used as the primary interface to view, query, and analyze logs generated by cloud services.

Logs Explorer enables centralized visibility into cloud activity, which is critical for monitoring, troubleshooting, and security investigations.

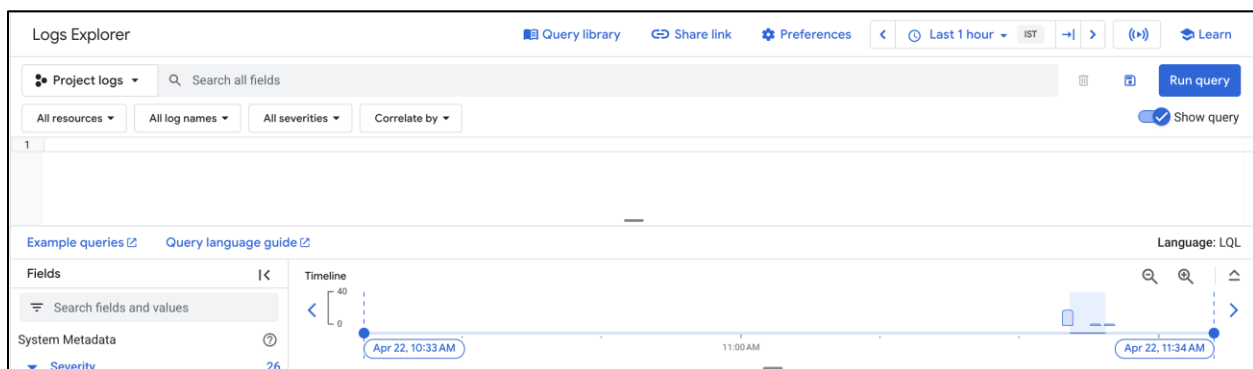


Fig. 1. Logs Explorer interface

Step 3: Viewing Cloud Log Entries

Multiple log entries generated by cloud services were observed, confirming successful log ingestion.

Observed logs included:

- System-level activity
- API operations

- Resource-level events

This step validated that cloud services continuously generate audit and operational logs that can be analyzed for security monitoring.

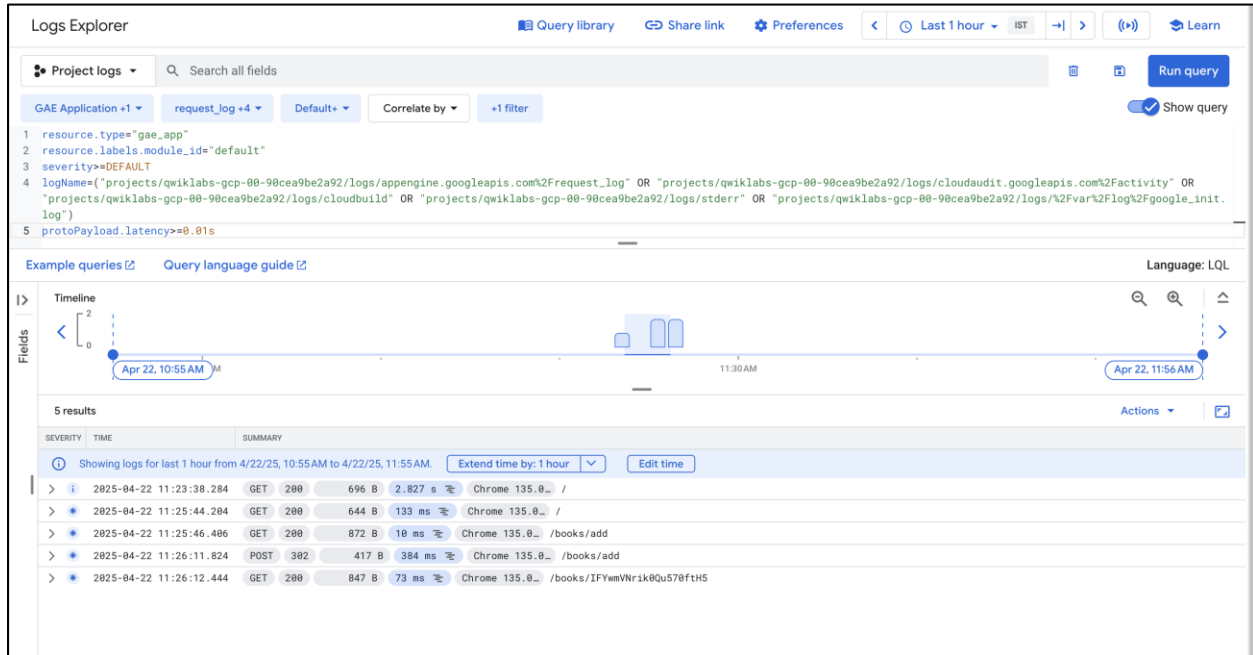


Fig. 2. Live log entries visible in Logs Explorer

Step 4: Filtering Logs by Compute Engine (VM)

Logs were filtered to display only **Compute Engine (Virtual Machine)** logs.

This filtering technique is essential for:

- Reducing log noise
- Isolating asset-specific activity
- Investigating VM-level behavior

This mirrors real-world scenarios where security analysts monitor specific assets for suspicious behavior.

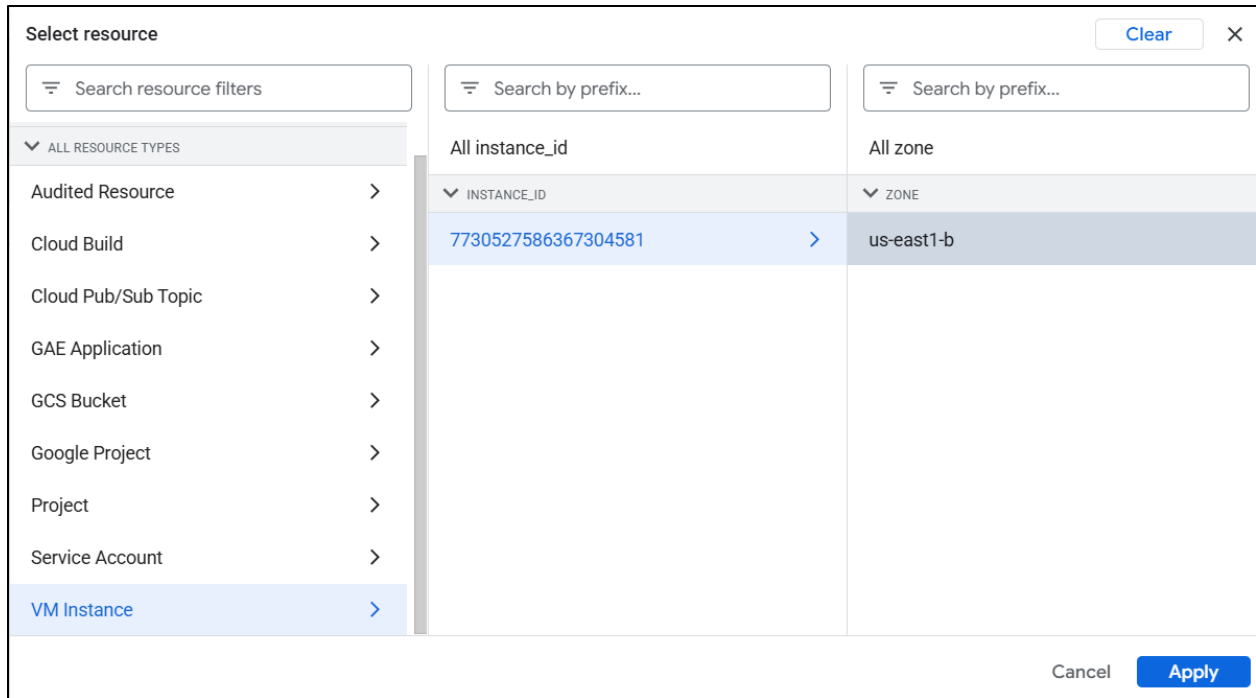


Fig. 3. Logs filtered to Compute Engine VM resources

Step 5: Severity-Based Log Filtering

Logs were further filtered using severity levels such as:

- **WARNING**
- **ERROR**
- **CRITICAL**

Severity-based filtering allows security teams to prioritize high-impact events and quickly identify potential security or operational issues.

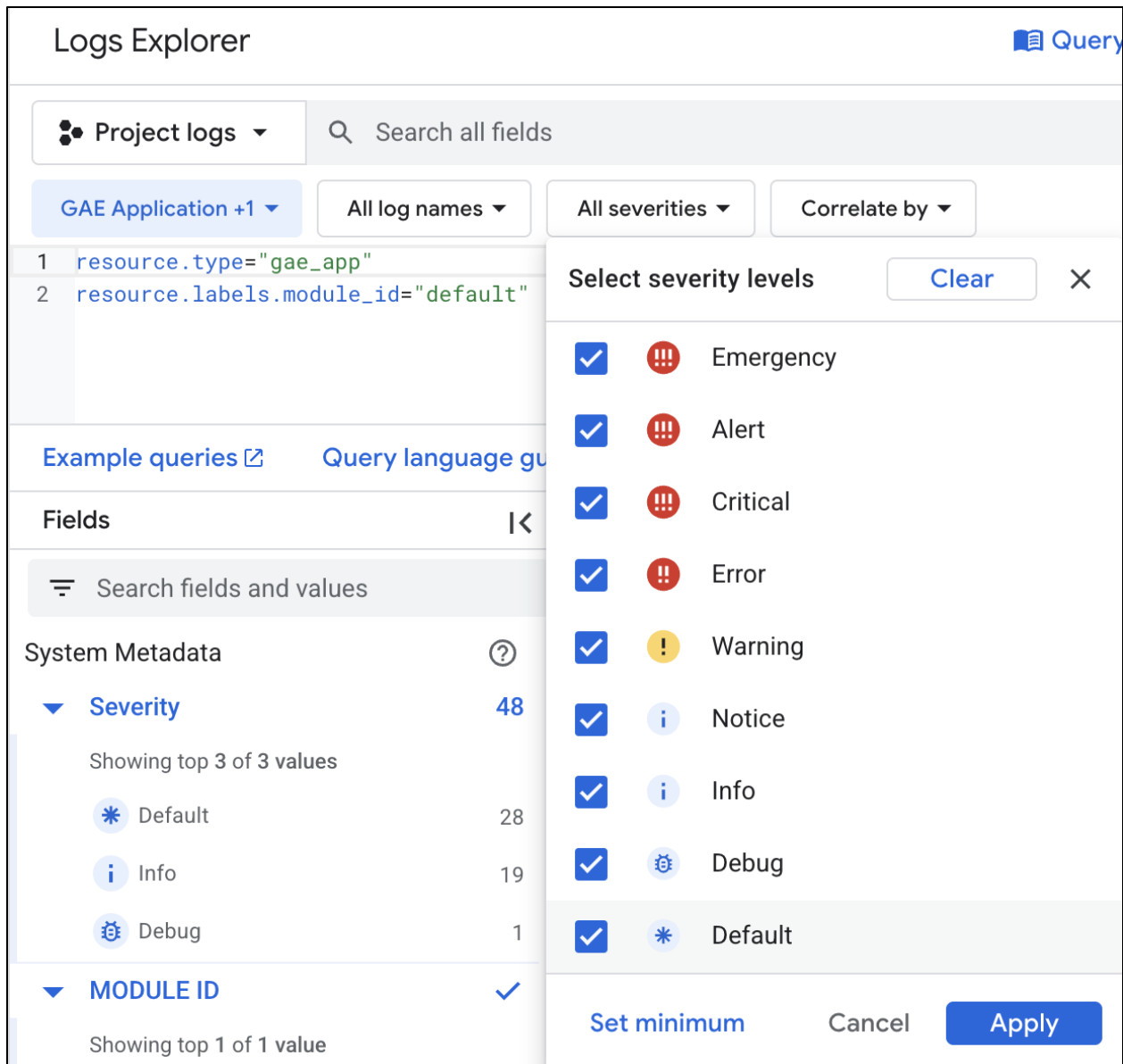


Fig. 4. Logs filtered by severity levels

Step 6: Log Analysis Using Log Queries

Log queries were used in **Logs Explorer** to analyze and filter cloud-generated logs relevant to security monitoring.

By applying structured log filters, logs were queried based on:

- **Resource type** (Compute Engine virtual machines)
- **Log severity levels** (WARNING, ERROR, CRITICAL)

- **Service-generated events**

This query-based approach helped isolate **security-relevant events** from high-volume logs, reducing noise, and focusing on actionable data.

Using log queries is a core practice in **SOC operations**, where analysts rely on filtering and querying logs to:

- Detect abnormal or suspicious activity
- Prioritize high-severity events
- Investigate infrastructure-level issues



Fig. 5. Log Query run

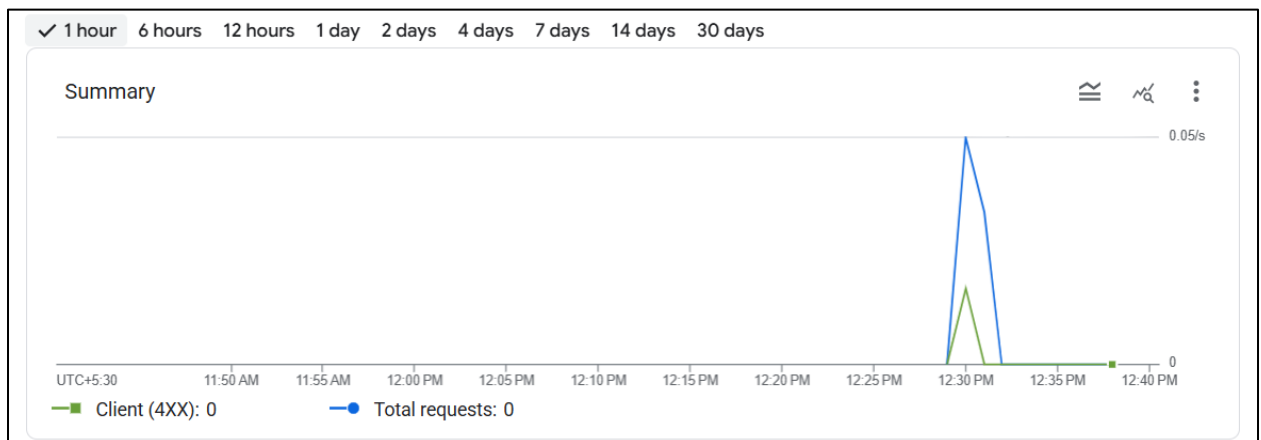


Fig. 6. Instances Activity

Security Analysis & Key Observations

- Cloud Logging provides centralized visibility into cloud activity
- Resource-based filtering helps isolate specific assets
- Severity levels assist in prioritizing potential incidents
- Detailed log metadata supports effective incident investigation
- Logs serve as critical evidence for cloud security monitoring

Project Outcome

- Successfully analyzed cloud-generated logs
- Applied resource-level and severity-based filters
- Investigated security-relevant log entries
- Gained hands-on experience with cloud security monitoring and logging